



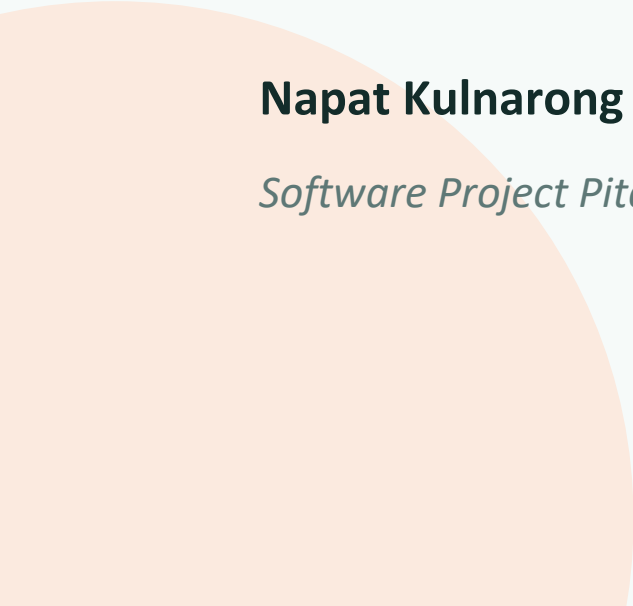
SOFTWARE PROJECT PROPOSAL

TORRENT

A centralized platform that tracks BMA software procurement (TOR) opportunities and proactively matches them to qualified vendors — a reversed job board for government software procurement.

Napat Kulnarong · Sethtatad Kijkanjanarat · Jirakorn Chaitanaporn

Software Project Pitching · Proposal



TORRENT (n.)

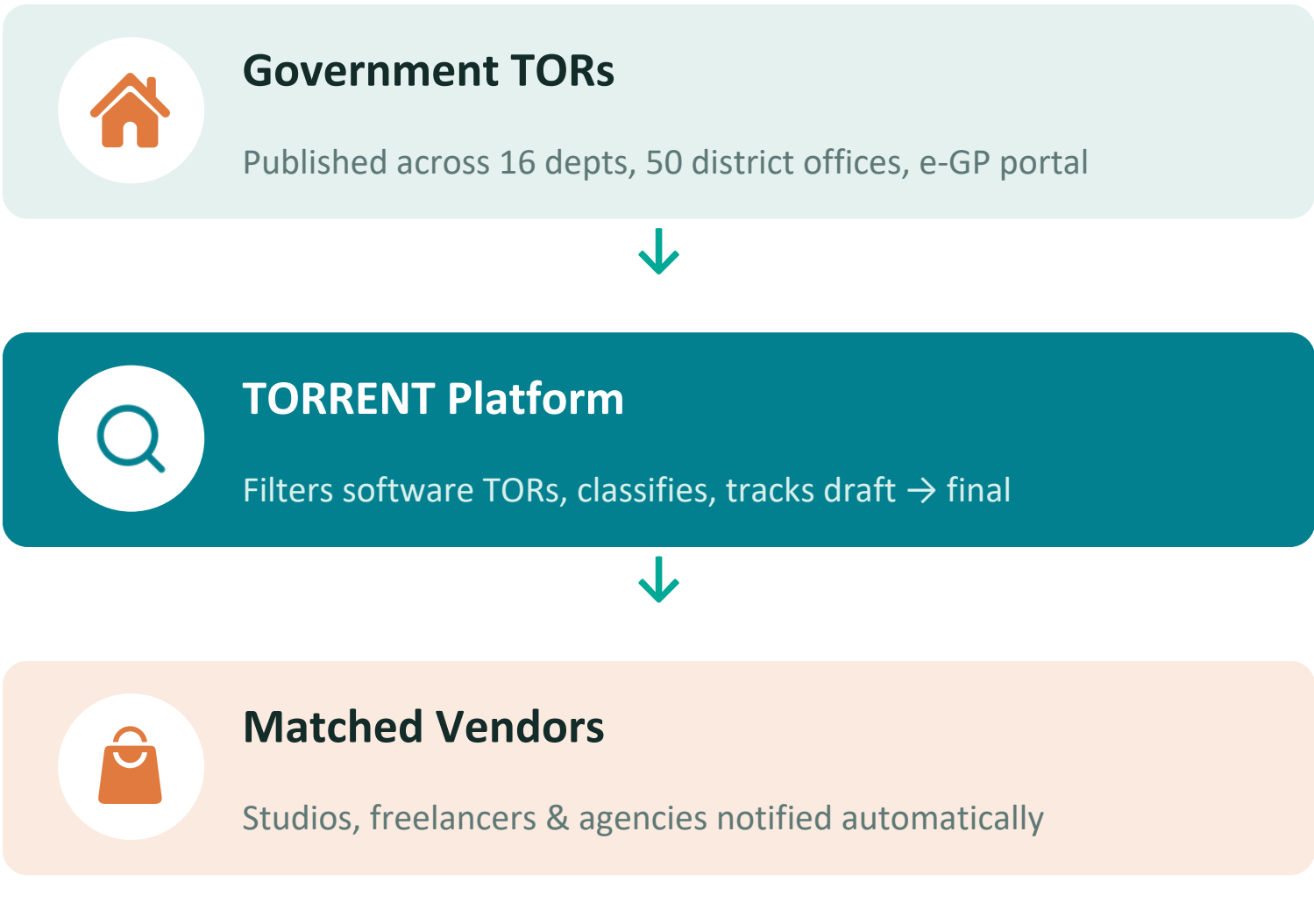
a fast and heavy flow of water, or a sudden, overwhelming outpouring of things like words or complaints.

A reversed job board for government software procurement

The BMA Software Procurement Platform is a centralized system that tracks software-related procurement opportunities (TORs) issued under the Bangkok Metropolitan Administration and proactively notifies qualified software vendors, studios, freelancers, and agencies about relevant opportunities.

Instead of vendors searching for work across dozens of scattered portals, TORRENT surfaces and pushes relevant government software TORs directly to them, from draft stage through publication.

HOW IT FLIPS THE MODEL



One portal. Zero filtering. No visibility.

16 departments + 50 district offices + hospitals & transit authorities — published on a single e-GP portal with no software filtering, no proactive alerts, no price history.



For Vendors

- No efficient way to discover relevant opportunities early
- No way to benchmark whether a bid is competitively priced
- Smaller studios & freelancers miss opportunities entirely
- Significant time spent manually checking dozens of portals



For BMA

- Procurement is fragmented across dozens of sub-units
- No easy way to demonstrate fair, competitive practices
- Limited visibility for smaller or newer vendors
- Larger incumbents dominate by default, not by merit

Four capabilities, one platform



TOR Ingestion & Tracking

Pulls procurement data from BMA's e-GP portal, tracking draft-stage and final published TORs separately, filtered for software-related projects.



Matching & Notification Engine

Analyzes each TOR's requirements, matches them against registered vendor profiles, and notifies qualified vendors at draft stage or publication.



Public Dashboard

Budget and pricing benchmarking across similar past projects — a mass comparison view of BMA software project costs that promotes transparency.



Differentiated Proposal / Showcase View

Surfaces how different teams approached the same TOR, helping vendors position themselves and giving BMA visibility into varied approaches.

Built to move fast, designed to prove trust

KEY VALUE PROPOSITIONS

- ✓ Early visibility into upcoming and draft TORs, not just final published ones
- ✓ A centralized, software-filtered view of a fragmented procurement landscape
- ✓ Public price and budget benchmarking across similar past projects
- ✓ Increased transparency and exposure for agencies procuring in good faith

TECH STACK

- **Next.js (App Router)**
Frontend + API surface
- **Node.js**
Scheduled ingestion worker
- **MongoDB Atlas**
Data storage · Atlas Search for semantic matching
- **Vertex AI (Gemini)**
TOR extraction, classification & vendor-matching

Ingestion runs as a separate scheduled worker rather than live in Next.js API routes.

Concept preview — full mockups in progress

High-fidelity wireframes are still being designed. Below is a conceptual sketch of the two primary views to orient sign-off discussion.

Public Dashboard — Budget Benchmarking

Software

Draft

Published

This Year

Avg. TOR Value: THB 1.2M



Similar past projects · cost per feature (illustrative)

Showcase View — Differentiated Proposals



Studio A — modular CMS approach

Approach summary · timeline · est. cost



Studio B — low-code rapid build

Approach summary · timeline · est. cost



Studio C — AI-assisted workflow

Approach summary · timeline · est. cost

10 weeks, four phases



Three developers, one workflow



Napat Kulnarong

Project Manager, Frontend, UX/UI design,



Sethtata Kijkanjanarat

Backend, CI/CD pipeline, Testing



Jirakorn Chaitanaporn

Backend, QA, AI integration

TEAM WORKFLOW



Daily updates

Short task posts ([nickname] – Task 1 – Task 2) to track momentum and to encourage daily work



Git process

No direct pushes to main — every change via PR, 1 approval required, branch deleted after merge (tracked via GitHub webhook)



Weekly sync

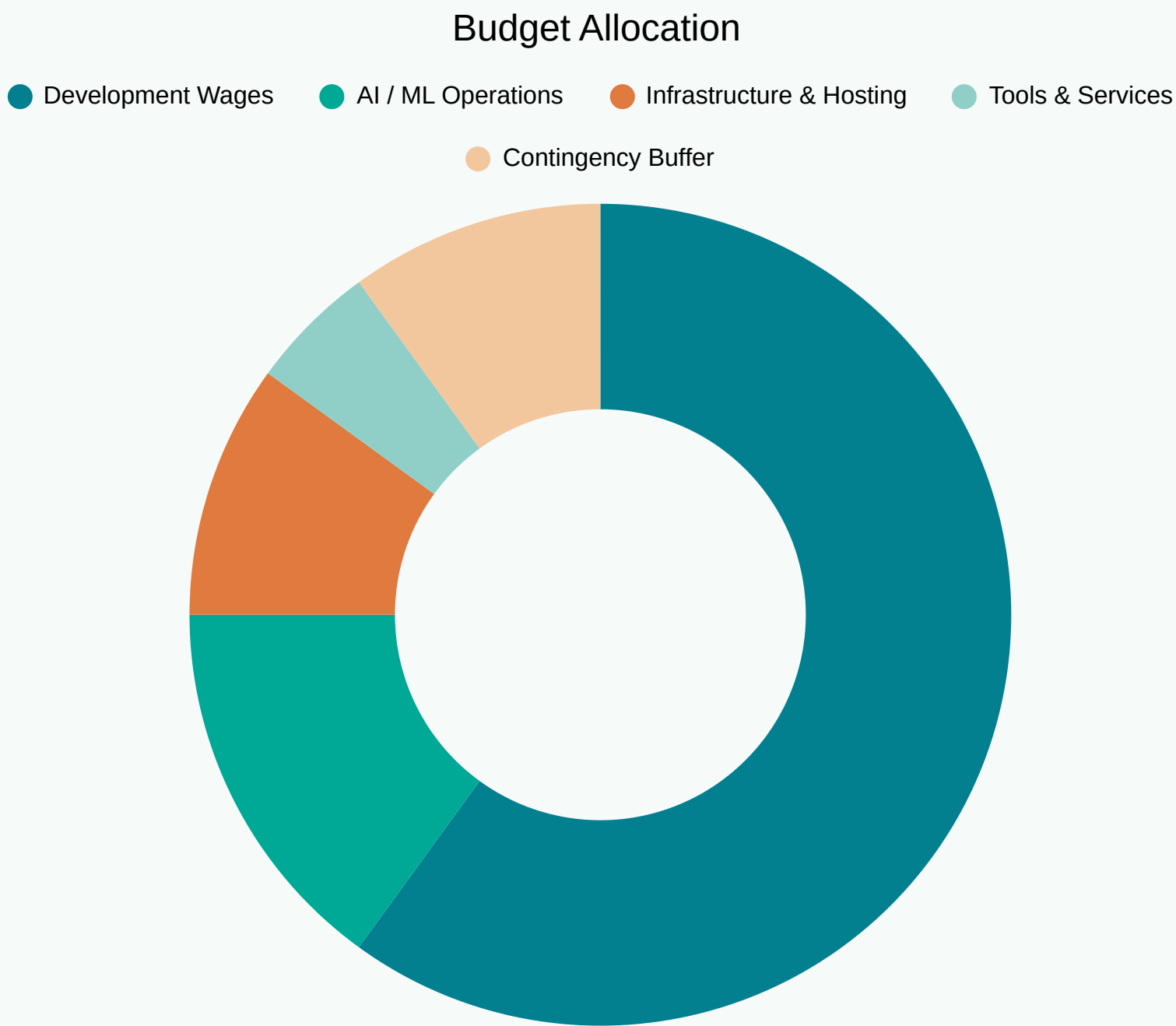
45–60 min recap after sync to realign & surface miscommunication early. This should take place on every Saturday or Sunday around 7.00 pm



Naming conventions

Commits: <type>(<scope>): <description ≤50 chars>.
Branches: <type>/<area>/<description>

1,000,000 THB full-project budget



BUDGET LINE ITEMS (THB)

Development Wages	600,000	Full-stack dev, UI/UX, system integration
AI / ML Operations	150,000	Vertex AI classification, parsing, matching
Infrastructure & Hosting	100,000	Year-1 hosting, DB, workers, SSL
Tools & Services	50,000	Email APIs, libraries, dev tool licenses
Contingency Buffer	100,000	Scope creep, scaling, API changes
TOTAL	1,000,000 THB	

Cost scales with entity scope: broad scope (50 district offices) means more ingestion volume, classification calls, and storage than the strict 16-department scope.